

Sathish Kumar Bani Sankar Jabade

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Profile

Master's student in Intelligent Robotics (M.Eng.) at Deggendorf Institute of Technology (Germany) with hands-on experience developing robotics perception pipelines on Linux. Built and integrated a YOLOv8-based QR victim-detection module for an AI-driven Search & Rescue TurtleBot3 system (ROS 2/Gazebo), including confidence gating, evidence logging, and map-frame victim reporting. Detail-oriented and process-driven, with prior automotive quality inspection experience and a strong focus on reproducible workflows and reliable results. Currently strengthening Python/C++ skills for robotics development.

Education

Master of Engineering in Robotics <i>Deggendorf Institute of Technology (DIT), Germany</i>	03/2025 – Present
Bachelor of Technology in Mechanical Engineering <i>PVKK Institute of Technology, India Grade: 2.0</i>	10/2021 – 05/2024
Diploma in Mechanical Engineering <i>PVKK Institute of Technology, India CGPA: 75.54</i>	06/2018 – 04/2021

Professional Experience

Automotive (4-Wheeler) Expert – Training <i>APSSDC – SIEMENS Center</i>	05/2023 – 06/2023 <i>Anantapur, India</i>
- Trained in diagnosis and maintenance of four-wheeled automobiles; gained practical workshop knowledge. - Developed technical competency relevant to real-world automotive industry requirements.	
Quality Control Inspector <i>KSH Automotive Pvt. Ltd</i>	06/2020 – 10/2020 <i>Penukonda, India</i>
- Performed quality inspection of metal parts for dimensional accuracy and surface finish using precision tools. - Conducted visual inspections, monitored production processes, and supported continuous improvement. - Completed final inspection before shipment; strengthened discipline, time management, and shop-floor teamwork.	

Projects

AI-Driven Search & Rescue with TurtleBot3 (ROS 2, SLAM, Nav2, YOLOv8) <i>Case study:- Autonomous exploration, and QR-victim detection.</i>	09/2025 – 01/2026
Goal: Build an end-to-end SAR pipeline that explores an unknown environment, maps it, avoids collisions in clutter, and reports victim locations with evidence. System stack: ROS 2 Humble on TurtleBot3 Waffle; <i>slam_toolbox</i> (occupancy grid), frontier-based exploration, Nav2 (global planning), TD3 RL local control, and a perception module for victim verification. My contribution (Perception): Implemented the complete YOLOv8 QR-detection pipeline running at 5 Hz, including confidence gating (trigger when confidence > 0.6) to reduce false positives. Victim reporting: Produced actionable outputs by logging timestamped evidence (image) and estimating victim coordinates in the map frame. ROS 2 integration: Packaged perception as a ROS 2 node/pipeline and published detections for downstream consumption (topics such as <code>/yolo/detections</code> and victim reporting interface).	
Vehicle Speed Estimation on Highway using Lucas–Kanade Optical Flow	08/2025 – 01/2026
- Implemented a computer-vision pipeline to estimate vehicle speed from highway videos using Lucas–Kanade optical flow to track feature-point motion between consecutive frames. - Added an optional perspective transformation (bird's-eye view) to reduce camera-angle distortion and improve motion measurement on the road plane.	

- Supported both input modes: prerecorded video files and live webcam feed; visualized tracked motion vectors directly on frames for debugging and validation.
- Converted tracked displacement over time into speed estimates and provided a CLI interface (e.g., `-input_file`, `-webcam`) for reproducible runs.

Fabrication of Solar Powered Mini Electric Tiller (Bachelor's Thesis)

01/2024 – 05/2024

- Designed and developed a cost-effective mini electric tiller to reduce manual labour using battery storage and solar energy.
- Focused on an eco-friendly and efficient alternative to conventional tillers.

Design, Analyse and Fabrication of Compact Foldable Electric Bicycle

07/2023 – 10/2023

- Designed a foldable electric bicycle for urban commuters as an eco-friendly mobility alternative.
- Integrated foldable design and dual-power functionality with a focus on sustainability for daily use.

Skills

- **Robotics:** ROS 2 (Humble), topics, message publishing (perception integration)
- **Perception:** YOLOv8 (Ultralytics), QR/victim detection, confidence thresholding
- **3D Estimation:** Bounding-box + depth alignment to estimate victim position in map frame, evidence logging
- **Simulation:** Gazebo
- **Programming:** Python (Ultralytics workflow)
- **OS:** Linux (Ubuntu)
- **CAD Tools:** AutoCAD
- **Computer Vision:** OpenCV, Lucas–Kanade Optical Flow, perspective transform, feature tracking

Publications

- **J. S. Bani Sankar Jabade**, Asst. Prof. B. Naveen Kumar, “*Fabrication of Solar Powered Mini Electric Tiller*,” International Journal of Advances in Engineering and Management (IJAEM), Vol. 6, Issue 8, Aug. 2024, pp. 274–284. DOI (as printed): 10.35629/5252-0608274284. PDF: ijaem.net link.
- **J. S. Bani Sankar Jabade**, B. Nikhil Yadav, J. Ravi Teja, *et al.*, “*Design, Analyse and Fabrication of Compact Foldable Electric Bicycle*,” International Journal of Advances in Engineering and Management (IJAEM), Vol. 6, Issue 8, Aug. 2024, pp. 480–488. DOI (journal listing): 10.35629/5252-0608475479. PDF: ijaem.net link.

Management and Leadership

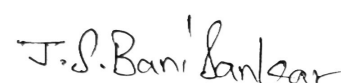
- Volunteer, Project-Expo-2024 at PVKKIT, Anantapur, India
- Student Coordinator, Aaviskhar-2023 at PVKK IT, Anantapur, India
- Organizer, Sport-tech-2022 (Zonal Level Sports Meet) at PVKKIT, Anantapur, India

Languages

Mother tongue: Telugu | **Marathi:** B2 | **English:** C1 | **German:** A1

Declaration

I affirm that all information provided is true and accurate to the best of my knowledge.



Kothmaibling, 20. February 2026